

Adrian Wenzen

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EDUCATION

Illinois Institute of Technology | Chicago, IL Expected **Summer 2027**
Bachelor of Science in Computer Science | GPA: 3.46/4.0
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Discrete Structures, Database Organization, Operating Systems, Probability & Statistics, Linear Algebra, Artificial Intelligence, Machine Learning, Software Engineering II

SKILLS

Languages: Python, SQL, R, JavaScript, TypeScript, C++, Java
Data Science & ML: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, TensorFlow, PyTorch
Data & BI: PostgreSQL, Microsoft SQL Server, Oracle SQL, Spark, dbt, Tableau, Power BI, ETL
AI & Tools: Cursor, Claude Code, Codex, Grok Build, Git, Docker, Jupyter Notebook, OpenAI API, Hugging Face, LangChain

CERTIFICATIONS

IBM Data Science Professional Certificate | IBM Jun 2026
Fundamentals of Quantitative Modeling | The Wharton School, University of Pennsylvania Nov 2025

TECHNICAL PROJECTS

Financial Asset Valuation & Risk Engine (J.P. Morgan Quantitative Research Job Simulation) | Personal Project Spring 2026
- Engineered quantitative engines for **Natural Gas Derivative Pricing** and **Consumer Credit Risk** assessment.
- Built a deterministic pricing model in **Python (Pandas)** to extrapolate seasonal trends and deployed a **Logistic Regression (Scikit-Learn)** classifier to score borrower risk.
- Uncovered a **\$120,000** seasonal arbitrage opportunity and quantified **\$8.23M** in Expected Loss (19.78% ratio) across a **\$41.6M** portfolio.

High Performance Options Pricer | Personal Project Fall 2025
- Engineered a low-latency C++17 engine to calculate Black-Scholes option prices and their corresponding risk sensitivities ("Greeks").
- Achieved a high-throughput benchmark of **3,090,517 options/sec** by optimizing all calculations into a single pass, avoiding redundant computation.
- Bridged the C++ core to Python using *pybind11* and validated mathematical correctness against known values with a *pytest* unit-testing suite.

Financial Market Analysis Dashboard | Personal Project Spring 2025
- Engineered an interactive dashboard using **Python** to analyze financial market data for a portfolio including **SPY (S&P 500 ETF)**, **BLK (BlackRock Stock)**, and **AGG (Aggregate Bond ETF)** over a **5+ year period (2020-01-01 to 2025-02-21)**.
- Implemented calculations for key financial metrics, including **annualized volatility (e.g., measured SPY volatility at 20.8%)** and **correlation matrices** to quantify relationships between assets.
- Designed the dashboard to provide an intuitive overview of **market trends, risk, and asset diversification**.

Machine Learning Stock Prediction Model | Personal Project Spring 2025
- Developed a **Classical Machine Learning** model (Random Forest) using **scikit-learn** and Python to predict >1% stock price movements based on technical indicators (SMA, RSI, MACD).
- Achieved 75% accuracy by training the model on a **large-scale time-series dataset** retrieved via yfinance.
- Analyzed feature importance to ensure **interpretability of results**, clearly explaining how technical indicators related to the **business objective** of forecasting price volatility.

Netflix-Inspired Backend | Personal Project Fall 2023
- Architected and implemented a **scalable database system** using **SQL**, optimizing queries for efficient content management and user data handling.
- Developed a **robust backend infrastructure** using **Python**, integrating authentication, content delivery, and **recommendation algorithms**.
- Optimized database indexing and query execution plans which **reduced data retrieval latency by 45%** during high-volume traffic simulations.

WORK EXPERIENCE

Software Engineer Intern | Transindonesia Network | Jakarta, Indonesia May 2026 - Jul 2026
- Built a **production-ready internal provisioning portal** to manage enterprise connectivity requests from customer intake, feasibility review, capacity validation, OSP/fiber checking, purchasing, approval, installation, activation, handover, and billing.
- Developed the system using **Next.js, TypeScript, PostgreSQL/Supabase, role-based authentication, workflow routing, task queues, document management, approval tracking, operational dashboards, CSV reporting, and automated activity logs**.
- Delivered an end-to-end operations platform supporting **8 internal user roles, 9 core modules, 10 provisioning stages, 50 standardized task outcomes, and 11 database entities**, improving execution visibility, reducing manual coordination, and creating a scalable system for future internal network service requests.

Co-founder & CTO | Loka | Jakarta, Indonesia Mar 2026 - Present
- Built a production-ready **community safety platform** for **Jakarta–Depok** covering live incident mapping, community reporting, **SOS**, official **BMKG alerts**, **CCTV integration**, and a **public waitlist with referral routing** at lokabyprox.com.
- Developed the stack using **SwiftUI, Mapbox Maps SDK v11, Firebase (Auth/Firestore/Hosting/FCM)**, **Firestore real-time listeners** with geohash indexing, **BMKG REST** ingestion, **HLS CCTV feeds**, and **vanilla JavaScript** with Firebase Web SDK v11.
- Delivered **v0.1.0 to TestFlight** with **10 internal testers** for device QA across map, GPS, reporting, SOS, and auth flows, validating the end-to-end product ahead of **Wave 1 launch**.

Student Consultant | RPA/IPA Automation | Chicago, US Aug 2022 - Dec 2022
- Selected for a **5-month consulting engagement**, analyzing **10+ business processes** and identifying opportunities for robotic and intelligent process automation.
- Evaluated **3 enterprise automation platforms** across **8 criteria**, including functionality, pricing, security, scalability, integrations, and implementation complexity.
- Designed and documented an **end-to-end automation workflow with 20+ process steps**, translating manual operations into an RPA/IPA solution and presenting implementation recommendations.